Attribute Closure (Closure set)

$R(A, B, C, D, E)$

FD: $\{A \to B, B \to C, C \to D, D \to E\}$

Exm 1

= contains set of attributes determined by A

CK | SK $A^+ = \{A, B, C, D, E\}$

$A \to C$
$A \to D$
$A \to A$
$A \to E$
$A \to ABCDE$

$B \to D$
$B \to E$
$B \to B$
$B \to BCDE$

$C \to E$
$C \to C$
$C \to CDE$

$D \to E$

$E \to E$

$AD \to BD$ (Aug)

$AD \to B$
$AD \to D$

$AD \to A, D$ (Reflex)

$AD^+ = \{A, D, B$

Easy: $AD \to A, D$ (Trivial

So,

$AD^+ = \{A, D$

$\{A \to B, B \to C, C \to D, D \to E\}$

$= \{A, D$

$A \to B$, so, $\{A, D, B\}$,

$B \to C$ so, $\{A, D, B, C\}$,

$D \to E$, so $AD^+ = \{A, D, B, C, E\} \to AD = SK$

Similar = $B^+ = \{B, C, D, E\}$

$CD^+ = \{C, D, E\}$

① SuperKey: Set of attributes whose closure contains all attributes of given relation

So, $A \Rightarrow SK \to$ 16 টা SK

Bcz, $A \Rightarrow SK$

So, $ABCDE = SK$

$R = (A, B, C, D, E)$

Find, $BCDE^+$ first

$BCDE^+ = \{B, C, D, E\}$, A নাই

যেহেতু BCDE মিলেও A নাই সুতরাং আলাদাভাবেও তারা SK নয়

③ CK: ~~যার থেকে কম~~ Minimal Super Key

সুতরাং, $A = CK$

SK $ABCDE^+$

বাদ করে নেই

$\times$ $ABDE^+ = \{A, B, D, E\}$

SK $ACDE^+ = \{A, C, D, E, B\}$

আরো ছোট

SK $ACD^+ = \{A, C, D, B, E\}$

$\times$ $AD^+ = \{A, D, B, E\}$

$AC = \times$

$CD = \{C, D, E\}$ $\times$

So, ACD is SK and CK

$R = (A, B, C, D, E)$

$FD - (A \to B,\ D \to E)$

Exm 2:

Easy Method of CK:

$SK = ABCDE$

~~ABCDE~~

Now. $A\cancel{B}CDE$ $\leftarrow [\because A \to B]$

$ACD\cancel{E}$ $\leftarrow [D \to E]$

SK $\longleftrightarrow$ ACD

$A \to B$ $D \to E$

Now, ACD কে আর ছোট করা যায় না,
তবুও এর সাবসেট গুলোকে চেক করা যায়

$\times \begin{cases} AC^+ \\ CD^+ \\ AD^+ \end{cases}$ $\quad A^+$, C^+, D^+

So, ACD = CK

Prime attributes = parts of candidate key

so, till now, prime attrib = A, C, D

If, no prime attri is on the right side of FD : $(A \to B,\ D \to E)$
then, no other candidate key.

Exam 3:

$R = (A, B, C, D)$

$FD: \{A \to B,\ B \to C,\ C \to A\}$

ABCD

AB CD [A→B]

A CD [A→B, B→C, ∴ A→C]

So CK = AD [আর ছোট হবে না]

কারণ চেক $\begin{cases} A^+ = \{ABC\} \\ \times \{D^+ = \{D\}\} \end{cases}$

Prime Attr $= \{A, D\}$ [FD = {A→B, B→C, C→A}]

prime Right side এ আছে

AD

CD = SK হবে

$\begin{cases} C^+ = \{C, A, B\} \\ \times \{D^+ = \{ \end{cases}$

∴ CD = CK

CD = [A→B, B→C, C→A]

Right

CD

BD = SK হবে

$\begin{cases} B^+ \\ D^+ = \end{cases}$

∴ BD = CK

BD [A→B B→C C→A]

Right

but AD is already found

So, all CK = 3 CK = {AD, CD, BD}